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SOUNDS AND COMMUNICATIONS OF THE YELLOW-BELLIED MARMOT (*MARMOTA FLAVIVENTRIS*)

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The present research is a study of the communications (the giving and receiving of information) of the yellow-bellied marmot. Auditory communications are of primary concern.

Warren (1935) mentions seven sounds of the yellow-bellied marmot (*Marmota flaviventris* (Audubon and Bachman)). The loud whistle he thinks expresses doubt or inquiry. The alarm call is a quieter 'coughed "keyuck" '. He reports a low whistling sound given from down in the burrow, expressing an inquisitive mood. The fourth call, given by feeding, playing, or sunning marmots, he does not describe. A variation of the whistle consisting of 'running a number of notes together in an undulating fashion' is the fifth call. The last two Warren mentions were given by a badly wounded marmot—a 'medium low "barking" cry' and a 'low whine'. Armitage (1962) in his article on the social behaviour of the yellow-bellied marmot mentions that the basic sound is a shrill whistle originating in the vocal chords. The alarm call seemed to him to be the 'basic note, but given higher and sharper'. Armitage also mentions a 'sound resembling a shriek' given by young at play or by a submissive adult when approached by a dominant male. The fourth call he mentions is a 'growl' given by marmots in their burrows. Seton (1929) observed a female yellow-bellied marmot move her less-than-a-week old litter. She dropped one young which 'uttered a loud shrill squealing' for 40 min until the mother retrieved it.

Methods and Materials

From June 1962 to June 1963, 89 days were spent observing yellow-bellied marmots in their natural environment. These rodents hibernate about 6 months of each year and are diurnal from mid-March to mid-September in Colorado. No less than 100 different marmots were observed—twenty-two of them intensively. Marmot communication in some form was almost always present in the natural situation.

Marmots were studied in two locations. One, an alpine area, was at an elevation of 11,700 ft on Taylor Peak, Gunnison County, Colorado (lat. 39°00′ N; long. 106°46′ W). The second, the foothills area, was at the border of the Great Plains and the Rocky Mountains at an elevation of 5240 ft in Larimer County, Colorado (lat. 40°35′ N; long. 105°10′ W).

The alpine colony had pikas (*Ochotona princeps*) living among the same rocks as the marmots, while cottontails (*Sylvilagus* sp.) shared the rocks of the foothills marmots. Coyotes (*Canis latrans*) and man were the major predators in both areas.

Observations were also made in Pitkin and Boulder Counties and in Rocky Mountain National Park, Colorado. Laboratory work was done at Colorado State University, Fort Collins, Colorado.

The sounds of the marmots were tape-recorded in the field and laboratory on a model W-610-EV Amplifier Corporation of America TransMagnemite battery-operated tape recorder. A tape speed of 15 in./sec was used for both recording and field playbacks. The microphone used was an Electro-Voice model 644 Sound Spot unidirectional microphone with a level frequency response of 50 to 12,000 cycles/sec. The net quality of the recordings was limited by the frequency response of the microphone. An Amplifier Corporation of America model DC-1S battery-operated amplifier and speaker was used with the recorder for field playbacks.

A Kay Electric Company model 661-A Sona-Graph, a sound spectrograph, was used to analyse the various sounds from 85 to 8000 cycles/sec. Wide band and HS analysis proved most satisfactory. For further analysis I used a Bruel & Kjaer model 2203 Precision Sound Level Meter, model 1613 Octave Filter Set, and type 4131 Condenser Microphone Cartridge. The calibration was done with a Bruel & Kjaer type 4220 Pistonphone microphone calibrator. The overall frequency range covered by the meter and filter set was from 22 to 45,000 cycles/sec. The investigations of the marmot sounds were limited to this frequency range.

The B & K Precision Sound Level Meter and accessories were used 14 in. from an adult male

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marmot within a model 1204 Industrial Acoustics Company Audiometric Testing Room while the marmot voluntarily whistled.

Also tested was the breathing associated with the vocalization. A pneumograph was placed around the posterior ribs of an etherized marmot, and connected to a recording physiograph. The marmot recovered from the visible effects of the ether and began to whistle. A continuous physiographic record was made and marked each time a whistle was emitted.

The postures and communicative behaviour patterns of marmots were studied in the field, from 16mm motion pictures, and from 35mm film transparencies. Individual marmots could be readily identified by the variation in the markings of their pelage—especially facial markings. I live-trapped the marmots at the foothills area to determine their sexes.

Sections of the recordings were played back to the marmots in the field after I first observed them for at least 15 min. In addition, I used a recording of the whistles of a hoary marmot (*M. caligata*) recorded by Charles Sutherland in Mt. Rainer National Park (cut 1, *Marmota* reel, Library of Natural Sounds, Cornell University). The resulting behaviour of the marmots listening to the playbacks was used as an indication of the communicative value of the sounds.

Results

I. Sound Production

The sounds of marmots are produced by both vocal and non-vocal means. The most obvious non-vocal sound of the marmot is produced by chattering teeth. The common vocal sound is the whistle. Motion pictures of marmots show that the whistling marmot opens its mouth widely for each sound, then partially closes the mouth before again opening it to emit the next whistle (Plate VIII). The teeth are not directly involved in the sound production of the whistle, which is produced during a rapid and forced exhalation (Fig. 1). While producing the whistle sound the marmot frequently views the stimulating object with monocular vision and faces about 90° to one side of this object (see Plate VIII).

II. Primary Whistle Motif

The yellow-bellied marmot sound most outstanding is the short, loud whistle (Plate IXA). This sound is the primary whistle motif. In contrast to the hoary marmot whistle (Plate IXB), the whistle of the yellow-bellied marmot is very short in duration—0·564 ± 0·073 SD (7) sec and 0·037 ± 0·006 SD (27) sec respectively. The harmonic structure of the sound in Plate IXA is typical of every yellow-bellied marmot whistle recorded. The fundamental begins at 3200 cycles/sec, rises to about 4000 cycles/sec, then rapidly drops in frequency. As shown in the sound spectrogram there are at least two harmonics above the fundamental. The first harmonic above the fundamental usually starts at 6400 cycles/sec, and like the fundamental, rises and falls in frequency. The data in Fig. 2, indicate that there are some harmonics in the ultrasonic frequencies.

While the relative intensity of the various portions of the whistle can only be estimated on the sound spectrogram, I was able to read intensity directly on the B & K sound level meter. The averaged results at a distance of 14 in. from the subject's mouth are shown in Fig. 2. The

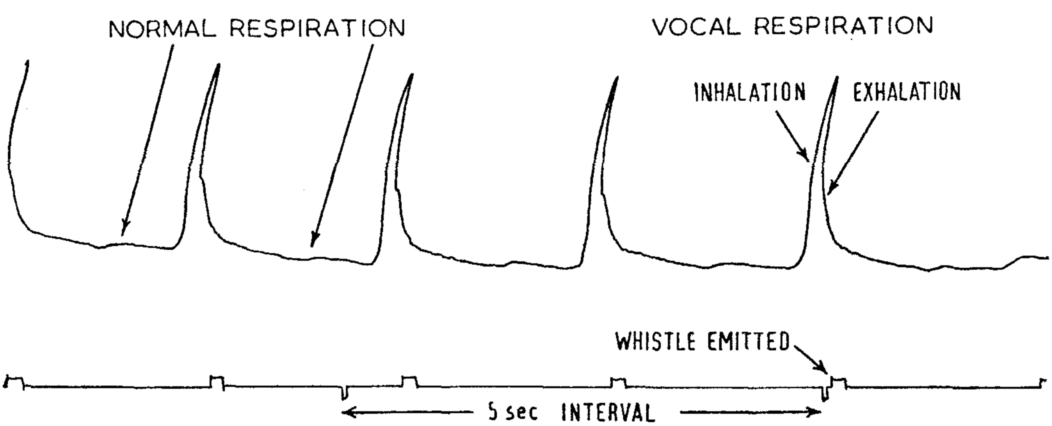
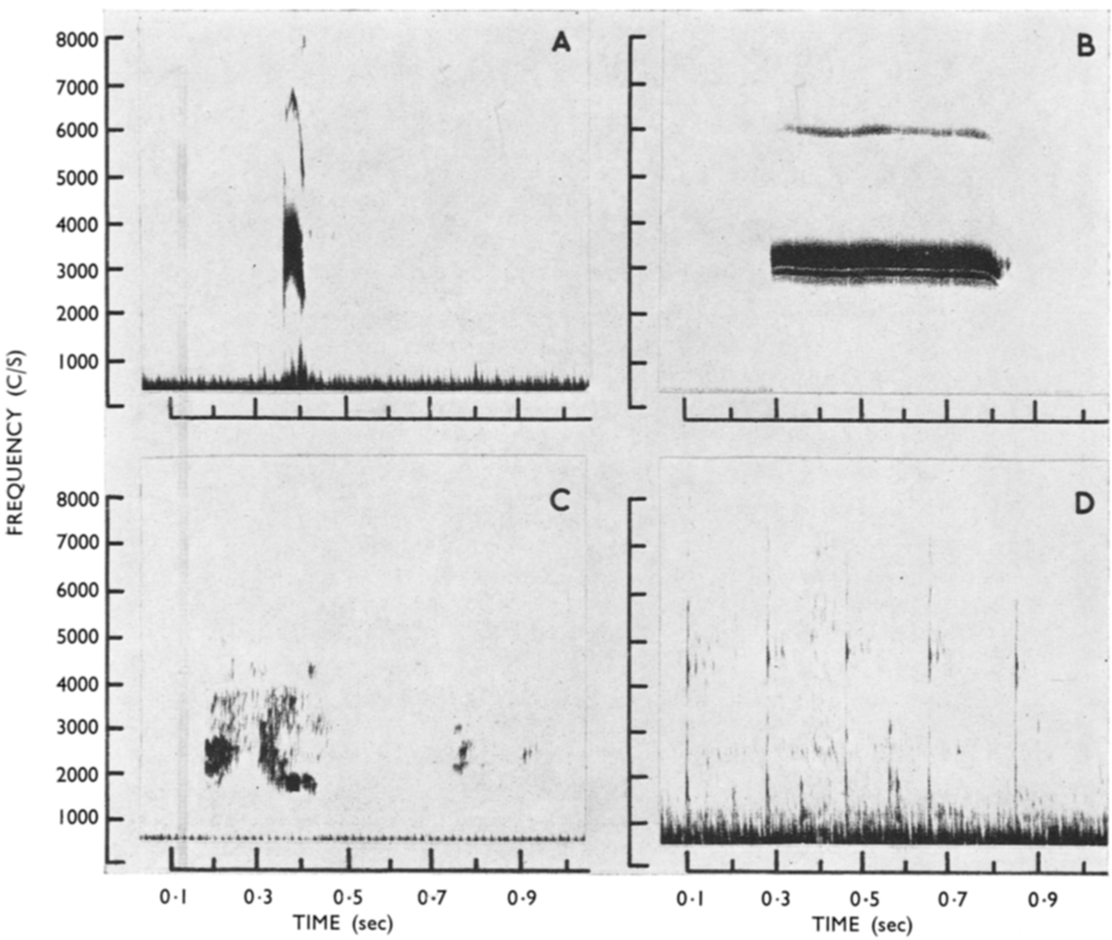


Fig. 1. Physiograph record of the respiration of a vocalizing marmot.

PLATE VIII

Marmota flaviventris while whistling.

PLATE IX



A. Primary whistle motif of *Marmota flaviventris*.
C. Screams.

B. *Marmota caligata* whistle.
D. Tooth chatter.

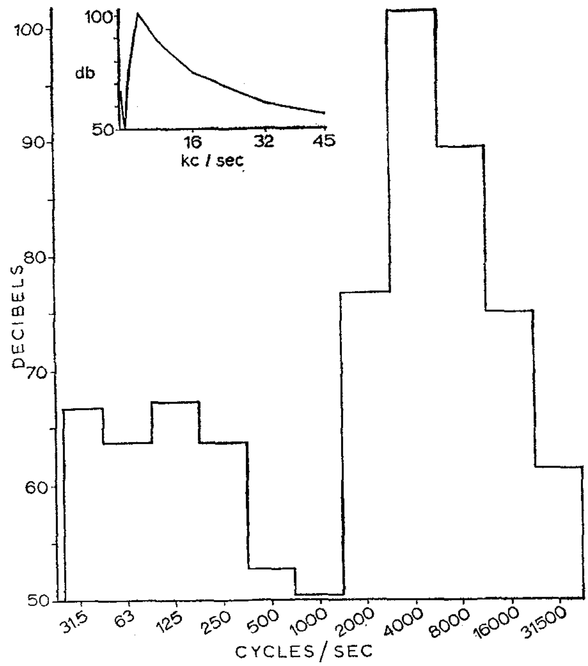


Fig. 2. Mean results of the analysis of yellow-bellied marmot whistles on a Bruel and Kjaer Precision Sound Level Meter.

loudest portion of the whistle was 101·3 dB between 3000 and 5000 cycles/sec. (For man, normal speech is at 60 dB; whereas 125 dB is painful.) The results also show that the intensity was 61·3 dB at the filter's centre frequency of 31,500 cycles/sec. At that octave band the frequency response of the instrument ends at 45,000 cycles/sec. From these data the upper frequency of the marmot whistle appears to be above 45,000 cycles/sec.

As noted, this whistle is the motif of most of the common sounds of the yellow-bellied marmot. With variations in rapidity, quality, and intensity of the whistle, the sounds indicate variations in stimuli and behaviour. However, the the sounds themselves may not be entirely functional for communications unless the animal listening is also using other senses within the same environment. Each of the following sounds I have heard used independently of the others by *Marmota flaviventris*.

(a) *Long-interval whistles.* The interval between whistles is an important variation in marmot sounds. These animals tend to repeat their sounds over and over again at a constant rate if the stimulus is also constant. The 'long-interval whistles' consist of the clear, short, loud primary whistle motif repeated three or more seconds apart. The interval most often is close to 10 sec. If the stimulation decreases or accommodation occurs the interval becomes longer, and if the stimulation increases the interval shortens and may grade into the 'short-interval whistles' discussed later.

These whistles are often used when the whistler is alerted but not greatly alarmed. The sound can be given from a prone, standing, sitting, or sitting-up posture by any marmot about 1 month old or older. Only one marmot in each 'coterie' gives the 'long-interval whistles' at the same time. It was possible to determine the location of the stimulus by noting where the whistles were coming from, for as the stimulating object left one location the marmot who was whistling there would soon be silent. The first marmot to see the intruder is usually the one that emits the whistles. Intruders such as deer do not easily stimulate marmots to vocalize in late summer, as marmots appear sluggish and not as wary as in the spring and early summer.

(b) *Short-interval whistles.* The 'short-interval whistles' are a series of whistles repeating the primary whistle motif with the interval between whistles less than 3 sec. One second or less intervals are frequent. There is no absolute separation in the 'short-' and 'long-interval whistles' in the natural situation. However, the marmots tend to respond to a stimulus with a high degree of excitement and often alarm, or they become alerted and temporarily cease their previous activity. I heard the 'short-interval whistles' given in a variety of situations, but in every instance the marmot giving the sounds appeared very excited or uneasy.

This series of whistles is often associated with stimulation by predators. A dog or a coyote is a strong stimulus for this type of sound. I observed one marmot give this series of whistles immediately after it ceased chasing a marten (*Martes americana*) out of the home range of the 'coterie' (Waring, 1965).

These whistles are given by weaned young, yearlings, and adults of both sexes, usually from a stiff-muscle sitting posture (see Plate VIII).

(c) *Quiet whistles.* The 'quiet whistles' are based on the primary whistle motif, but are altered mainly in decreased intensity. The drop in intensity allows the non-vocal sounds of the gular region to be heard. Therefore, the 'quiet whistles' are made up of weak, high-pitched whistles and gagging or choking sounds. The result is more of a noise than a clear whistle. The interval between sounds is usually not over 10 sec nor much less than 1 sec.

The 'quiet whistles' were given by marmots which saw me near their burrow as they began to come out and also by apprehensive captive marmots. If the animal was stimulated more, the sounds gradually changed to the 'short-interval whistles'.

These whistles are given from sitting and sitting-up postures by weaned young, yearlings, and adults of both sexes.

(d) *Accelerando whistles.* Another series based on the primary whistle motif is the 'accelerando whistles', which begin with the loud, clear whistle motif. The interval between the first few sounds may be more than 3 sec, but with each succeeding sound the interval shortens. After only six whistles the interval may decrease until the marmot is giving as many as five sounds per second. At the more rapid rate the quality of the whistle motif seems to be lost, and the sound appears to be a rapid vocal chatter. Normally the marmots inhale before each whistle sound is emitted; however, during the 'accelerando whistles' the most rapid sounds are made from one inhalation. After that breath has expired, the sound series may stop, change to the 'tooth chatter', or change to the 'short-interval whistles'.

Marmots use a brief form of the 'accelerando whistles' as agonistic sounds. This consists of the last part of the accelerating whistle series. The most rapid whistles are often followed by the 'tooth chatter'. Adult marmots give this call from sitting and prone postures. Captured adults could be induced to give these agonistic sounds by annoying them, and would often lunge at the side of the cage during the rapid ending of the call to attack the annoying object.

(e) *Barking whistles.* If a marmot is vocalizing while running, the whistles have a slight barking sound. The sound is based on the primary whistle motif; however, it has additional frequencies making the sound a noisy whistle. At times the sound duration is slightly longer than the primary whistle motif. The interval between sounds is often irregular. Males and females can give the 'barking whistles' at least after 1 month of age.

The marmots wander from their home burrow while feeding or showing investigative behaviour. If they see something approaching they often give the 'barking whistles' while running to a rock for another look. The whistles may cease en route, or they continue as 'short-' or 'long-interval whistles' when the marmot reaches its observation rock. A marmot observed chasing a marten emitted the 'barking whistles' during the entire chase (Waring, 1965).

(f) *Mono-emphatic whistle.* The last marmot whistle sound is the single, extra loud whistle—the 'mono-emphatic whistle'. It is given by weaned young, yearlings, and adults of both sexes. This whistle has the duration, pitch, and quality of the primary whistle motif; however, the intensity is increased over any of the other sounds of the marmot.

The marmots observed making the 'mono-emphatic whistle' were always starting to move from a stationary position to their burrow. The more common situation where the marmot uses the extra loud whistle is when the animal begins to run from imminent danger. The marmot often has only enough time to emit the one sound; yet, when it watches the danger slowly approach, the 'short-interval whistles' may precede the 'mono-emphatic whistle'.

III. Screams

The yellow-bellied marmots produce the 'screams' vocally by prolonging sounds similar to the whistles. The 'screams' vary in duration, and they are usually longer than the primary whistle motif (Plate IXC). Typically the sound is made up of many frequencies and lacks good harmonic structure. What looks to be the fundamental undulates between 1500 and 2500 cycles/sec. The intensity of the entire cry undulates, also.

These sounds were heard only rarely and were usually given by submissive marmots when chased by a dominant.

IV. Tooth Chatter

Another sound the marmots produce is the 'tooth chatter'. By anteroposterior and vice-versa movements of the lower jaw the marmot alternatively hits the distal and proximal surfaces of the upper incisors with the lower incisors. The result is a rapid, non-vocal, clicking sound. The 'tooth chatter' is usually preceded by the threatening ending of the 'accelerando whistles'.

Captured adult marmots that were annoyed by a foreign object violently charged with chattering teeth and bumped the side of the cage with their nose and forepaws. The spectrogram of the 'tooth chatter' (Plate IXD), shows the sound occurs at all frequencies up to about 8000 cycles/sec. The duration is less than 0·01 sec.

V. Other Means of Communication

Marmots do not communicate by sound alone. Sight is very important in their communications, and is probably used in most social interactions. Little sound if any is used in sexual, care-giving, or care-soliciting behaviour. Sight obviously is important in communicating intention movements and postures.

A method of communication using sight as well as olfactory and tactile means is the behaviour designated by Armitage (1962) as the 'greeting'. I have observed this behaviour between young marmots, adult marmots, and also between mother and young. The 'greeting' consists of one marmot approaching usually from the side or front of the other. The approaching marmot usually with its tail curved and held up slightly moves its head alongside of the other. The two marmots then mutually touch tactile hairs and smell along the side of the other's head to the buccal tactile hairs and sometimes as far as the ear. This behaviour may then be followed by one or both smelling and often touching the anal region of the other. Possibly the anal glands are functional in the final part of this 'greeting' behaviour.

Pikas (*Ochotona princeps*) quickly respond to the warning whistles of the yellow-bellied marmot in places where they share the same habitat. Though the marmots are usually more observant of the entire surroundings than their neighbours, marmots themselves may sometimes rely on auditory and visual interspecies communication.

The hoary marmot sounds caused no response when played back to the yellow-bellied marmots.

Discussion

The results of the present research indicate that the many sounds produced by the yellow-bellied marmot are used to communicate with their own and other species. Marmots do not ignore the alarm sounds of another marmot nearby; they react quickly. Many communications, however, need not be 'acted upon' and are received solely for the information derived from the communication (e.g. information received to enable recognition of an individual).

A sudden contraction of the abdominal muscles forcing air through the larynx is responsible for most of the vocal sounds of the yellow-bellied marmot. A marmot does not usually emit an initial sound when it first becomes alerted. Usually marmots are aware of danger long before it is near, and do not become tensed suddenly enough to produce any noticeable sounds. Generally they watch briefly in a rigid posture before uttering the first sound. It is not unusual for a marmot to flee silently when confronted with sudden danger.

Situations where danger is present seem to be communicated in several ways by marmots. When a marmot quietly but hastily retreats into a burrow, the marmot's neighbours appear to understand that something is wrong just by this action. If a threatening object is seen approaching slowly at a distance or a nearby object seems to have very little threatening appearance, the sounds given will usually be the 'long-interval whistles', which serve an alerting function for the neighbouring marmots. If the stimulus is stronger, as when a threatening object is nearby, the sounds given will be more rapid—the 'short-interval whistles', indicating alarm and causing neighbouring marmots to run for their home burrows. The 'accelerando whistles' also can be given when a disturbing object is rapidly approaching the whistling marmot from a distance.

Possible danger is, also, indicated by the 'quiet whistles' and the 'barking whistles'. The 'quiet whistles' are often given when the threatening object is first seen motionless near the marmot. The 'barking whistles' indicate that the marmot is running and the interval between sounds would indicate the degree that the marmot was stimulated. If the danger were close the intervals would be shorter.

The yellow-bellied marmots do not repeat any of the sounds from one to another. There is only one sound-producing marmot at a time in each 'coterie', and that marmot alone continues to emit the sounds until the stimulus is gone.

It now appears evident that marmots have a number of ways to communicate the presence of danger.

Alarm sometimes may be indicated when a marmot gives the 'mono-emphatic whistle', but it may also be used when the situation seems peaceful, and there is no apparent cause for alarm. Therefore, it seems the sound is mostly an indication of the vocalizing marmot's descent into its burrow.

The 'all clear' of the yellow-bellied marmots is a visual communication, not auditory. The lack of sound may, however, indicate that danger is no longer present. When danger has passed the marmot that whistled, or other nearby marmots, resume their previous activities. It is by this resumption of their activities that they appear to communicate an 'all clear'.

The marmots observed did not appear disturbed in any way when an eagle or other large bird flew overhead. Thus, I would not expect an 'aerial predator warning' to exist among marmots.

Agonistic sounds, such as scolding or threatening, are given as the 'accelerando whistles' and the 'tooth chatter'. When a marmot appears to be scolding another marmot the sound is the rapid accelerating whistles. If the marmot must face imminent danger it threatens by giving the 'accelerando whistles' followed by the 'tooth chatter'. The marmot at this time is tensed, ready to fight, and may charge without hesitation.

The 'scream' of the yellow-bellied marmot is not necessarily produced because the marmot is suffering pain. When a marmot appears to be afraid of harm to itself, the distress or submissive sound is the 'scream'. However, 'screams' can be from either a fearful or a pleasurable experience. Knowledge of the existing situation may be necessary to understand the use of the 'screams'.

Many marmot communications are done by body postures and movements. The marmot, though capable of some movements of its pinnae, does not appear to use the ear movements to communicate. The marmot tail, however, is rather obvious. As a feeding or wandering marmot waddles up a slope the tail waves behind in a large arc. If the marmot is running from a threatening situation or is running rapidly to its observation rock the tail no longer waves but is carried down and nearly motionless.

I did not observe any gustatory communications; however, olfactory and tactile communication was observed. The ritual of the 'greeting' appears to be a communication using visual, olfactory, and tactile stimulation. Recognition and seeking acceptance by the other seem to be the main functions of the 'greeting'.

Interspecies communication does exist with marmots and their neighbours as was noted with the pikas.

The type of communication and whether or not it is given will be based entirely on each particular situation and is difficult to predict. Their sounds are often modified slightly by each individual. Some marmots are more prone to be easily excited and begin vocalizing readily; others do not. 'Long-' or 'short-interval whistles' may be given for more than 30 min by an individual that appears to be in a nervous emotional state in a seemingly normal and peaceful environment. The entire meaning of a communication is lost without knowledge of the individual variations and knowledge of the existing situation within the environment.

Table I. Summary of the Sounds of the Yellow-bellied Marmot

Name of sound	Characteristics	Common function
Primary whistle motif (PWM)	Formant at 4000 cycles/sec	Motif of all whistle series
Long-interval whistles	PWM series with intersound interval of 3 sec or more	Alert
Short-interval whistles	PWM series with intersound interval less than 3 sec	Alarm
Quiet whistles	Low intensity PWM series with gular noise	Alert or alarm
Accelerando whistles	PWM series with progressive decrease in inter-sound interval	Threat or alert
Barking whistles	Variable interval PWM series given while running	Threat or alert
Mono-emphatic whistle	Single loud PWM	Descending into burrow
Screams	Undulating squeals	Fear or pleasure
Tooth chatter	Clicking incisors	Threat

Summary

The sounds and communications of the yellow-bellied marmot were studied from June 1962 to June 1963, during the seasons these marmots were not hibernating. Auditory communications were of primary concern. The study areas varied in altitude from 5240 ft to about 12,000 ft. The two main study areas were in Gunnison and Larimer Counties, Colorado. The behaviour of the marmots was observed in the field and from films; their sounds were tape-recorded on a battery-operated tape recorder of high quality. Recordings were played back to the marmots in the field with the recorder and a battery-operated amplifier and speaker. The sounds were analysed on a sound spectrograph. A sound level meter was used in the laboratory to measure the intensity of the sounds of a captive marmot.

Yellow-bellied marmots older than 1 month of age and of either sex can produce at least three main types of sounds—'whistles', 'screams', and 'tooth chatter'. The whistle sounds are modified by marmots to form six different calls. Some sounds may be used for more than one purpose, and purposes, such as giving alarm, are communicated by more than one means. Furthermore, any sound given within a 'coterie' is not repeated by any other member of that social group. Interspecies communication between marmots and the animals around them seems to exist.

Auditory, visual, tactile, and olfactory stimulations are used by marmots for communicating. Most communicating situations involve some form of visual stimulation, such as tail positions.

Yellow-bellied marmots do not seem to have an 'aerial predator warning' call nor an 'all-clear' call.

Very little communication can occur unless the receiving animal is familiar with the animal giving the communication, and is also aware of the situation existing within that animal's environment.

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